Welfare Analysis of a Forest-Residue SAF Supply Chain under Endogenous Demand and Price Uncertainty

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Motivation: a six-fold gap

- U.S. **SAF Grand Challenge**: 3 B gal/yr by 2030
- 2025 reality: 0.5–0.7 B gal/yr at $\sim$\$9/gal, with \$5.68/gal premium over Jet-A
- Closing the gap requires a coordinated buildout of pretreatment depots and biorefineries
- Southeastern U.S. has the largest commercially recoverable forest-residue supply

	B gal/yr
Goal (2030)	3.0
2025 actual	0.6
Gap	**2.4**
Our chain capacity	3.3

Three weaknesses in existing supply-chain MILPs

Existing residue-to-SAF MILPs (You 2012, Sharma 2013, Lin 2014, Roni 2017) treat the problem as cost-minimization at a fixed selling price.

Three economic weaknesses:

1. Exogenous SAF price assumes away **market clearing**.
2. Fixed-price formulations cannot decompose project value into **CS, PS, and externality components** — the decomposition needed for policy counterfactuals.
3. Deterministic NPV models offer **no risk metric** despite high payoff variance in an emerging fuel market.

Pigou 1920; Holland-Hughes-Knittel 2009; Rockafellar & Uryasev 2000.

Four contributions

1. **Endogenous SAF demand**
   Linear $P_{\text{SAF}}(Q) = a - bQ$ calibrated to the SAF Grand Challenge target. SAF treated as buyer-segmented (ESG offtakes, 45Z, LCFS), not perfect substitute for Jet-A.
2. **Recourse-cost upper bound by candidate evaluation**
   LP-only sweep over wait-and-see solutions; 18.8 min vs. failed extensive form. Headline: \$0.925 B/yr.
3. **Welfare-loss CVaR via Pareto enumeration**
   Rockafellar-Uryasev CVaR over $L_s = W_s^{\text{WS}} - W_s^{\text{HN}}$ across 20 candidates. Two undominated plans, \$7.1 M/yr risk premium.
4. **County-level aggregation**
   Shapely-based area shares; matches hex-level GAMS within $\pm 0.15\%$ on cost; nodes 14k $\rightarrow$ 443.

Model overview: two-stage stochastic MILP

First-stage binaries (location/capacity, scenario-independent):

- $\text{DvD}_{d,a}$ — depot d active in ash class a
- $\text{DvB}_{r,a'}$ — biorefinery r active in ash class a'

Second-stage flows (per scenario):

- $x_{h,d,a}$: residue hex $\rightarrow$ depot
- $z_{d,a,r,a'}$: pretreated depot $\rightarrow$ refinery

Scale (county-aggregated):

- $|H| = 443$, $|D| = |R| = 483$, $|A_d| = |A_r| = 10$
- $\sim$700,000 continuous, $\sim$9,660 binaries per scenario

Path B: endogenous demand replaces fixed PSAF

Path A (old): fixed PSAF, demand floor + slack import, minimize cost − revenue.

Path B (new): drop slack, replace fixed PSAF with $P(Q) = a - bQ$, maximize welfare:

$$W = \int_0^Q P(q)\,dq - C(Q) = I_{\text{CS}} - C$$

implemented via SOS2 breakpoints on $I_{\text{CS}}(Q) = aQ - bQ^2/2$.

Why "buyer-segmented"? Under perfect substitutes ($\text{MC}_{\text{SAF}} \approx \$5.50 > \$3.50 = P_{\text{fossil}}$), the social planner picks Jet-A and the chain is never built. ESG mandates, 45Z, and LCFS create demand for *certified* SAF.

Calibration: $(Q^*, P^*) = (3\,\text{B}, \$5.97/\text{gal})$

Anchor: long-run 2030+ scenario meeting SAF Grand Challenge.

Elasticity $\varepsilon_{\mathbf{SAF}} \in \{-0.3, -0.5, -1.0\}$, baseline -0.5.

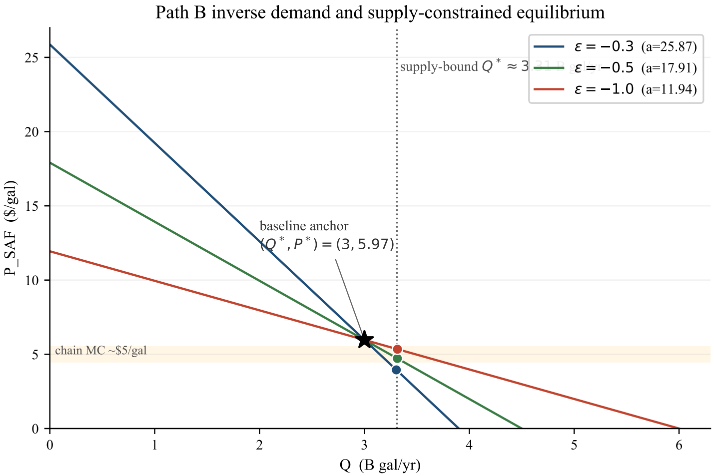
Three pieces of evidence:

- Pass-through: $\varepsilon_{\text{passenger}} \cdot s_{\text{fuel}} \approx -0.22$
- Brazilian ethanol: -1.5 (Cardoso et al. 2019)
- SAF more elastic than baseline jet fuel (fossil substitute available)

ε	a (\$/gal)	$Q_{\max}$ (B)
-0.3	25.87	3.9
-0.5	17.91	4.5
-1.0	11.94	6.0

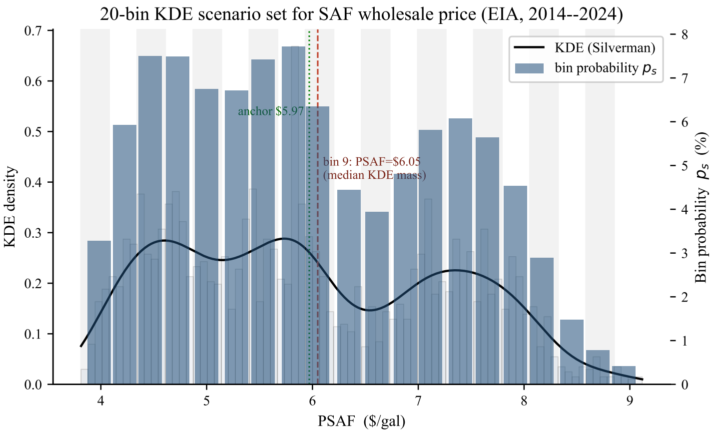
Linear inverse demand $P = a - bQ$ with $b = -P^*/(\varepsilon Q^*)$.

Inverse demand and supply-constrained equilibrium



All three elasticity equilibria fall on the supply ceiling $Q^* \approx 3.31\,\mathrm{B}$; P^* slides down the demand curve below the chain MC band $\Rightarrow$ **negative producer surplus**.

KDE scenario set: SAF wholesale-price uncertainty



20-bin Gaussian KDE on EIA Gulf-Coast jet-fuel wholesale 2014–2024, Silverman bandwidth. Bin 9 ($6.05/gal) sits at median KDE mass and at the topological-expansion threshold.

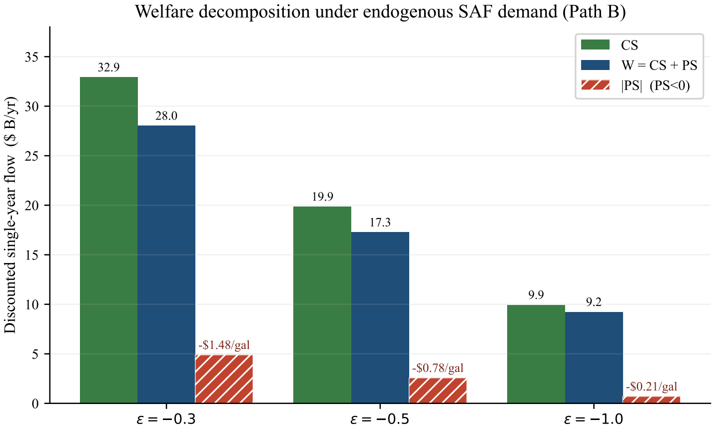
Result 1: Path B welfare equilibrium

ε	Q^* (B)	P^* (\$)	CS (\$ B)	PS (\$ B)	W (\$ B)	dep	ref
-0.3	3.304	3.95	32.92	-4.89	28.03	248	34
-0.5	3.314	4.72	19.86	-2.57	17.27	248	34
-1.0	3.315	5.34	9.94	-0.70	9.23	248	34

Three observations:

- Q^* and topology essentially invariant in $\varepsilon \Rightarrow$ **supply-constrained regime**
- P^* tracks the demand curve at the supply-bound Q^*
- PS robustly negative; per-gallon gap scales as $1/|\varepsilon|$ (\$0.21–\$1.48/gal)

Result 2: welfare decomposition by elasticity



CS-to-|PS| ratio rises with demand steepness: 14.2 ($\varepsilon = -1.0$) to 6.7 ($\varepsilon = -0.3$). In a supply-constrained regime, any policy that doesn't relax the supply ceiling **primarily redistributes between CS and PS**.

Result 3: 45Z bridge

The negative-PS gap is a quantitative reference for the IRA Section 45Z producer credit:

$$\text{Credit}_{45Z} = \$1.75 \times \max\left(0,\ \frac{50 - \text{CI}}{50}\right), \quad \text{CI in kg } CO_2\text{e/MMBtu}.$$

For low-CI Fischer-Tropsch forest-residue (CI $\approx$ 5–11 kg/MMBtu):

	Planner gap	45Z credit
$\varepsilon = -0.3$	\$1.48/gal	
$\varepsilon = -0.5$	\$0.78/gal	\$1.38–\$1.56/gal
$\varepsilon = -1.0$	\$0.21/gal	

Caveat: alignment is qualitative. Planner gap is $AC(Q^*) - P^*_{\text{planner}}$; 45Z compensates competitive private producers. Formal mapping in companion paper.

The stochastic question

What is the optimal first-stage capacity choice when the price scenario is unknown?

Two SP quantities:

- $\mathbb{E}[\mathrm{PVC}_{\mathrm{WS}}] = \mathbb{E}_s[\min_y \mathrm{PVC}(y, s)]$ (wait-and-see)
- $\mathbb{E}[\mathrm{PVC}_{\mathrm{RP}^*}] = \min_y \mathbb{E}_s[\mathrm{PVC}(y, s)]$ (recourse)

Standard SP inequality:

$$\mathbb{E}[\mathrm{PVC}_{\mathrm{WS}}] \leq \mathbb{E}[\mathrm{PVC}_{\mathrm{RP}^*}] \leq \min_c \mathbb{E}[\mathrm{PVC} \mid y_c^*]$$

Problem at our scale:

- Extensive form fails (barrier stalls at numerical noise floor)
- Integer Benders feasible but expensive (master keeps all 9,660 binaries plus 20 cuts/iteration)

Methodology: candidate-evaluation upper bound

Quick-Win procedure:

1. Take each wait-and-see solution y_c^* as a non-anticipative candidate
2. Fix binaries to y_c^*; collapse to LP
3. Re-evaluate over all 20 KDE scenarios via parametric LP-only sweep
4. Compute $\mathbb{E}[\text{PVC} \mid y_c^*] = \sum_s p_s \, \text{PVC}(y_c^*, s)$

Bound: $\mathbb{E}[\text{PVC}_{\text{RP}^*}] \leq \min_c \mathbb{E}[\text{PVC} \mid y_c^*]$.

Wall time: 400 LPs in 18.8 min (vs. failed extensive form). Each LP $\approx 0.8\,$s after first build with primal-simplex warm start.

Uses the same LP-tight subproblem oracle as Benders, but evaluates it at a hand-picked candidate set rather than via master iteration.

Quick-Win UB result

$\mathbb{E}[\mathrm{PVC}_{\mathrm{RP}^*}] \leq \$0.925\,\mathrm{B/yr}$

Achieved at $c^* = 20$ (highest-PSAF wait-and-see plan, 278 dep / 44 ref).

No tight lower bound. Legacy $\mathbb{E}[\widehat{\mathrm{PVC}_{\mathrm{WS}}}] = \$1.541\,\mathrm{B/yr}$ was computed at 5%-gap MIPs and carries upward bias of $\geq \$616\,\mathrm{M/yr}$. Tightening would require re-solving 20 sweep MIPs at $\leq 0.5\%$ gap ($\approx$ 100–200 CPU-h).

Methodology takeaway: when the extensive form fails, candidate evaluation gives a constructive UB at $1/50$ wall time, portable to other large-scale recourse problems.

Risk-aware analysis: setup

For each scenario s with $a_s = a_{\text{base}} + (\text{PSAF}_s - \overline{\text{PSAF}})$:

- W_s^{WS}: full Path B optimum at s
- $W_s^{\text{HN}}(y_c)$: Path B with binaries fixed to y_c, evaluated at s
- $L_s(y_c) = W_s^{\text{WS}} - W_s^{\text{HN}}(y_c)$

Risk-aware planner:

$$\min_{y_c \in \mathcal{C}} \,(1-\lambda)\,\mathbb{E}_s[L_s] + \lambda\,\text{CVaR}_\alpha(L_s), \quad \lambda \in [0,1].$$

Approach: enumerate all 20 wait-and-see candidates as $\mathcal{C}$; evaluate $(\mathbb{E}[L_c], \text{CVaR}_{0.05}(L_c))$ for each.

20-candidate Pareto frontier

Plan (PSAF)	dep / ref	$\mathbb{E}[L]$ \$ M/yr	$\text{CVaR}_{0.05}$ \$ M/yr	Note
bin 6 (\$5.28)	225 / 31	**92.7**	435.6	E[L] best
bin 8 (\$5.82)	227 / 31	99.8	**427.8**	CVaR best
bin 9 (\$6.05)	248 / 34	97.7	723.4	central; *dominated*
bin 1 (\$3.99)	193 / 27	472.2	1,909.5	smallest WS
bin 20 (\$8.94)	278 / 44	1,161.5	2,529.0	largest WS

Two Pareto-undominated candidates; the deterministic Path B equilibrium plan (bin 9, 248/34) is *dominated* on both axes by bin 6.

Wall: 85 min total (one Path B WS sweep + 20 HN sweeps).

Risk premium and the deterministic-vs-stochastic gap

Risk premium (RN $\to$ fully CVaR-averse):

$$\Delta\mathbb{E}[L] = +\$7.1\,\mathrm{M/yr}, \quad \Delta\mathrm{CVaR}_{0.05} = -\$7.8\,\mathrm{M/yr}$$

Switching threshold $\lambda^* \approx 0.48$.

Deterministic plan is dominated:

- Bin 9 over-builds 23 dep + 3 ref beyond bin 6
- Pays off only in supply-constrained high-PSAF tail ($\sim$25% prob)
- In low-PSAF mass, capex sunk

Det.-vs-stoch. $\mathbf{CVaR_{0.05}}$ gap = \$300 M/yr: cost of using deterministic-central plan in a stochastic decision problem.

Sensitivity

Bandwidth (Silverman scale):

mult.	QW UB	$\mathbb{E}[L]$	$\text{CVaR}_{0.05}$
$0.5\times$	\$0.867 B	\$100.2 M	\$716 M
$1.0\times$	\$0.934 B	\$ 97.3 M	\$724 M
$1.5\times$	\$1.026 B	\$ 93.8 M	\$732 M

- Path B CVaR robust to $\pm 1\%$; QW UB to $\pm 8\%$
- SOS2 grid (interior-Q): 21 vs. 31 breakpoints agree on W within 0.86%
- All three ε pick same 248/34 topology in deterministic problem

Policy implications

1. **Supply ceiling drives chain scale**, not elasticity. Levers that don't relax the residue ceiling primarily redistribute CS$\leftrightarrow$PS. Levers that do (feedstock procurement, residue-recovery, depot densification) have first-order welfare leverage.
2. **Production-credit gap aligns with 45Z magnitude**. Per-gallon planner deficit \$0.21–\$1.48 is comparable to current 45Z (\$1.38–\$1.56 for low-CI FT residue). Alignment is qualitative; formal mapping in companion paper.
3. **Deterministic plans are dominated under uncertainty**. The median-KDE-bin equilibrium choice is strictly worse than the bin-6 plan on both axes, with a \$300 M/yr CVaR gap.

Limitations and extensions

Three limitations:

- **SAF-as-separate-market** defended on policy grounds, not from direct empirical SAF-demand estimate
- **Linear inverse demand** is local approximation
- **No carbon externality**; Pigouvian τ + 45Z step deferred to companion paper

Extensions:

- Solve joint Rockafellar-Uryasev MILP at low gap
- Re-solve sweep at $\leq 0.5\%$ gap to tighten EVPI
- Hybrid candidate plans (non-WS) for richer Pareto frontier
- Pathway commit (FT vs. ATJ vs. pyrolysis); BT23 full-stream supply

Summary

What we did:

- County-aggregated 2-stage stochastic MILP for SE U.S. forest-residue $\rightarrow$ SAF
- Endogenous SAF demand + Marshallian welfare decomposition
- Candidate-evaluation UB on recourse cost (vs. failed extensive form)
- Welfare-loss CVaR via 20-candidate Pareto enumeration

Headline numbers:

- Break-even: **\$5.83/gal**; topological threshold **\$6.05/gal**
- Recourse-cost UB: **\$0.925 B/yr**
- Welfare equilibrium ($\varepsilon = -0.5$): **\$17.27 B/yr**; PS gap \$0.78/gal
- Risk premium: **\$7.1 M/yr**
- Det.-vs-stoch. CVaR gap: **\$300 M/yr**

Thank you.

Questions?

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